Projections, Extent, and Resolution

HES 505 Fall 2025: Session 8

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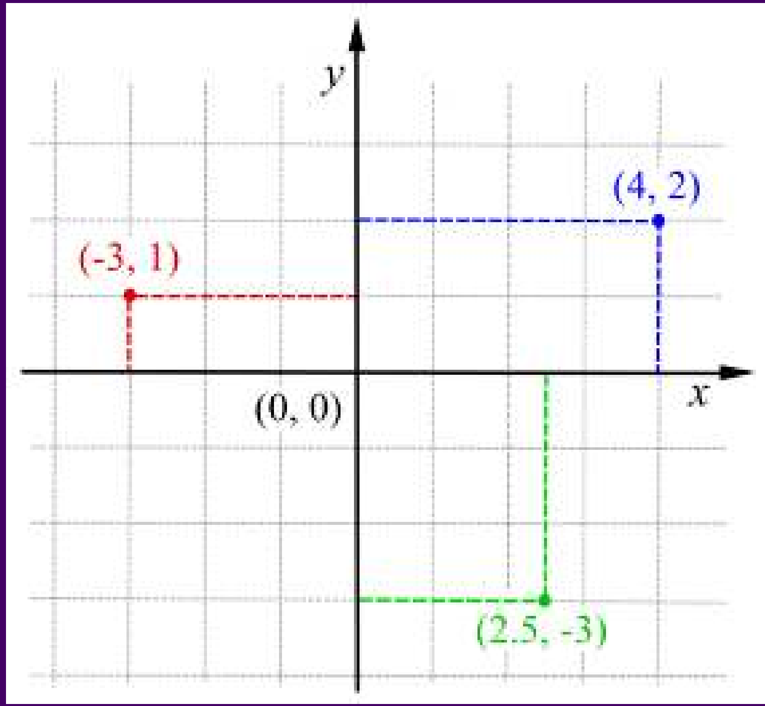
Describing Absolute Locations

- **Coordinates:** 2 or more measurements that specify location relative to a *reference system*

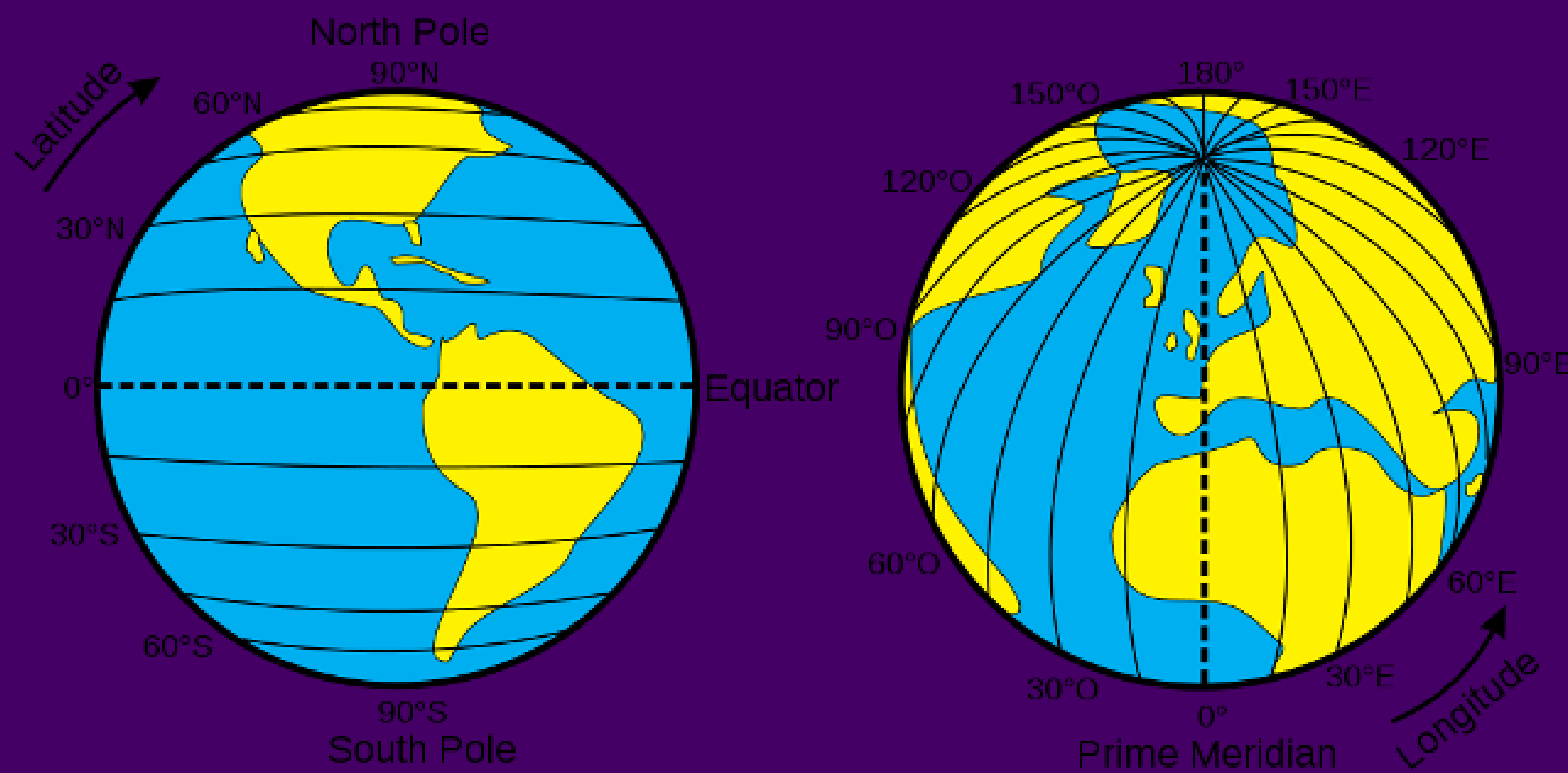
- Cartesian coordinate system
- *origin (O)* = the point at which both measurement systems intersect
- Adaptable to multiple dimensions (e.g. *z* for altitude)



Cartesian Coordinate System

Locations on a Globe

- The earth is not flat…



Latitude and Longitude

Locations on a Globe

- The earth is not flat…
- Global Reference Systems (GRS)
- *Graticule*: the grid formed by the intersection of longitude and latitude
- The graticule is based on an ellipsoid model of earth's surface and contained in the *datum*

Global Reference Systems

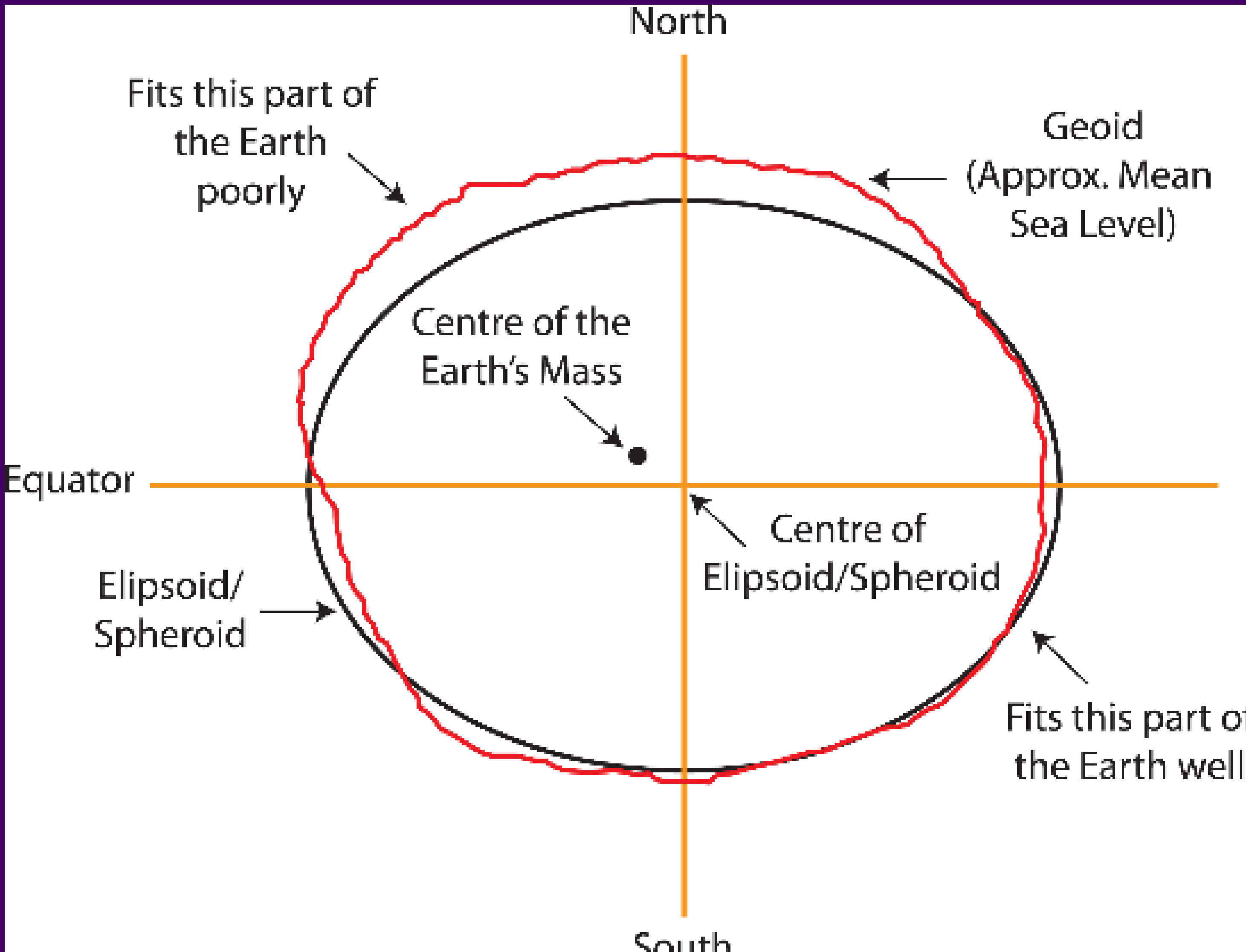
The *datum* describes which ellipsoid to use and relies on angular measurements to describe location on a sphere

- Geodetic datums (e.g., `WGS84`): distance from earth's center of gravity
- Local data (e.g., `NAD83`): better models for local variation in earth's surface

Global Reference Systems

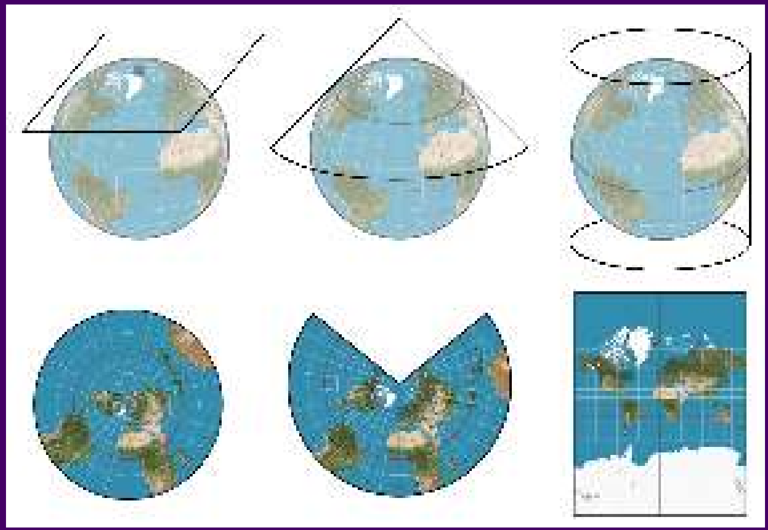
- Great for *precise location*
- Great for *worldwide communication*
- Not great for measurements (e.g., distance, area)
- Inconsistent accuracy

Global Reference Systems



The World Is Not Flat

- But maps, screens, publications, and Cartesian Coordinates are…
- **Projections** describe *how* the data should be translated to a flat surface
- Rely on 'developable surfaces'
- Strictly: the mathematical function to transform the globe into the developable surface.



Developable Surfaces

Projections

It’s just maths…

\[\begin{aligned} n &= \tfrac{1}{2}(\sin \phi_1 + \sin \phi_2) \\[4pt] C &= \cos^2 \phi_1 + 2n \sin \phi_1 \\[4pt] \rho(\phi) &= \frac{\sqrt{C - 2n \sin \phi}}{n}, \quad \rho_0 = \frac{\sqrt{C - 2n \sin \phi_0}}{n} \\[8pt] x &= \rho(\phi)\,\sin\!\big(n(\lambda - \lambda_0)\big) \\[4pt] y &= \rho_0 - \rho(\phi)\,\cos\!\big(n(\lambda - \lambda_0)\big) \end{aligned} \]

\[\begin{aligned} \phi &= \text{latitude} \\[4pt] \lambda &= \text{longitude} \\[4pt] \lambda_0 &= \text{central meridian} \\[4pt] \phi_0 &= \text{latitude of origin} \\[4pt] \phi_1, \phi_2 &= \text{standard parallels} \end{aligned} \]

Coordinate Reference System

- Contains the full recipe for translating Earth to coordinates
- Begin with a **datum** (model of spheroid Earth)
- **Projection** - the math for flattening (e.g., Albers, Mercator, etc)
- **Parameters** - the projection center, parallels, etc

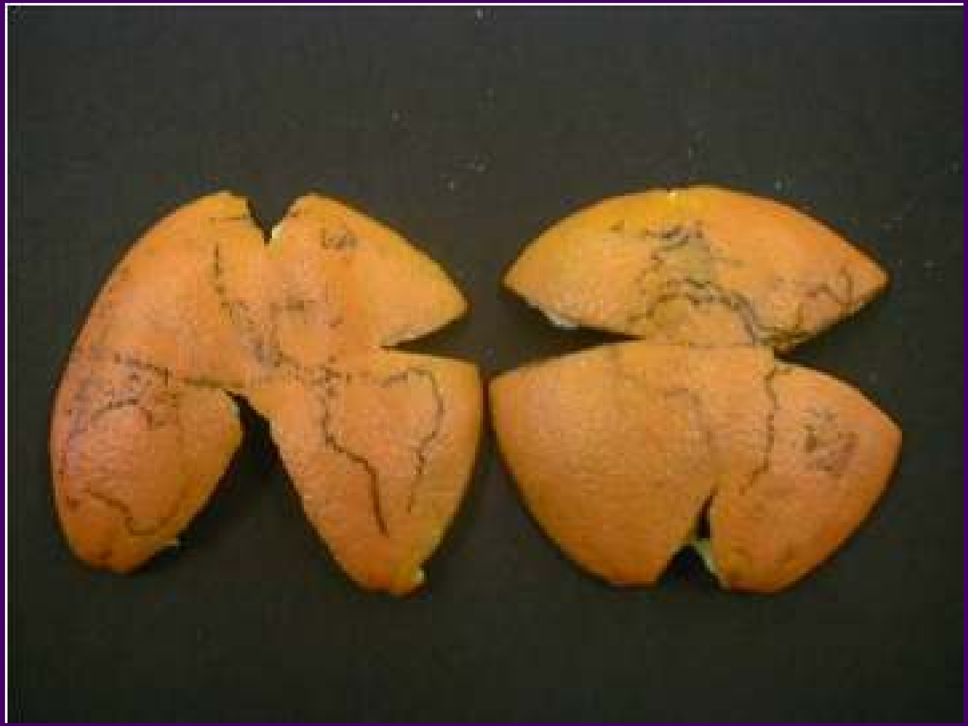
Coordinate Reference Systems

- Some projections minimize distortion of angle, area, or distance
- Others attempt to avoid extreme distortion of any kind
- Includes: Datum, ellipsoid, units, and other information (e.g., False Easting, Central Meridian) to further map the projection to the GCS
- Not all projections have / require all of the parameters

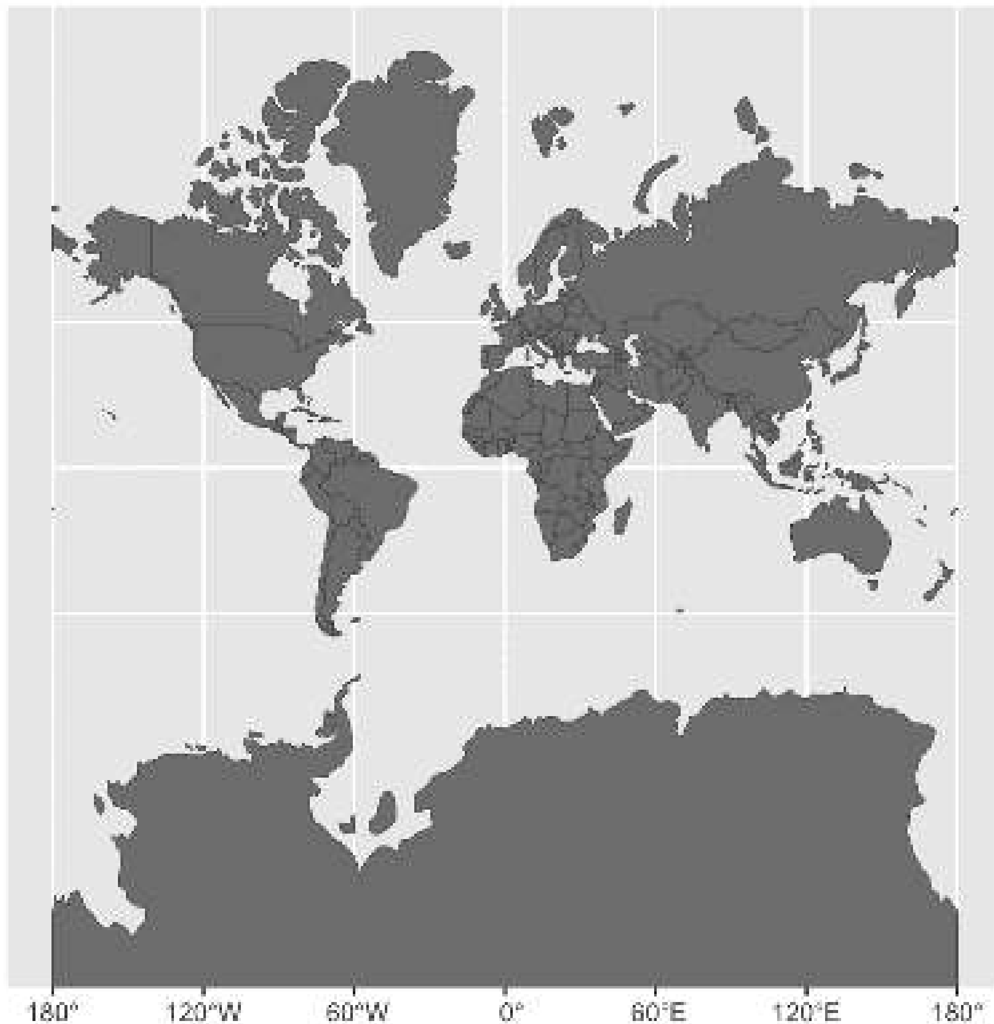
The Orange Peel Analogy

A datum is the choice of fruit to use. Is the earth an orange, a lemon, a lime, a grapefruit?

A projection is how you peel your orange and then flatten the peel.

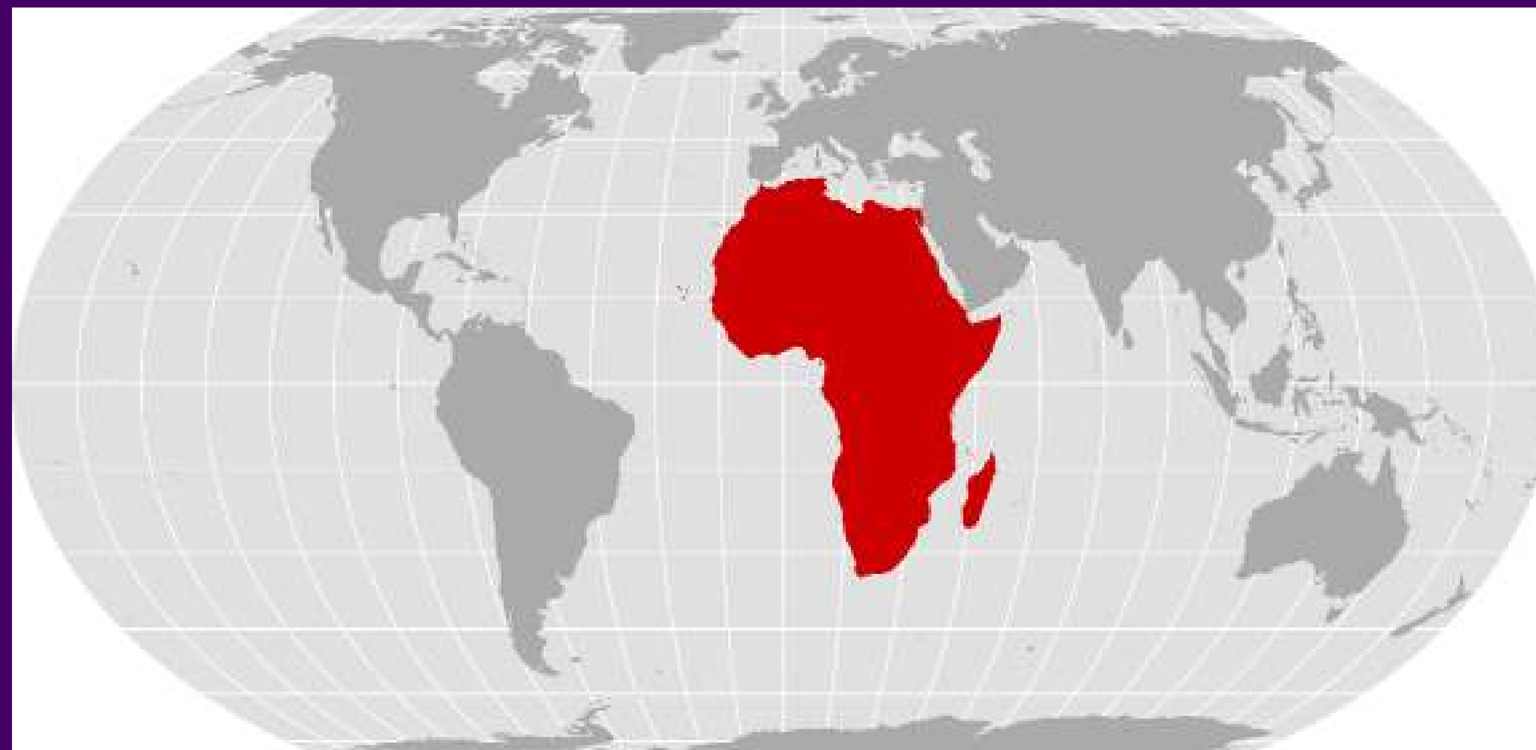


Projections are Political



Projections are Political

Projection necessarily induces some form of distortion (tearing, compression, or shearing)



Mercator vs. Equal Earth projection from Correct the Map

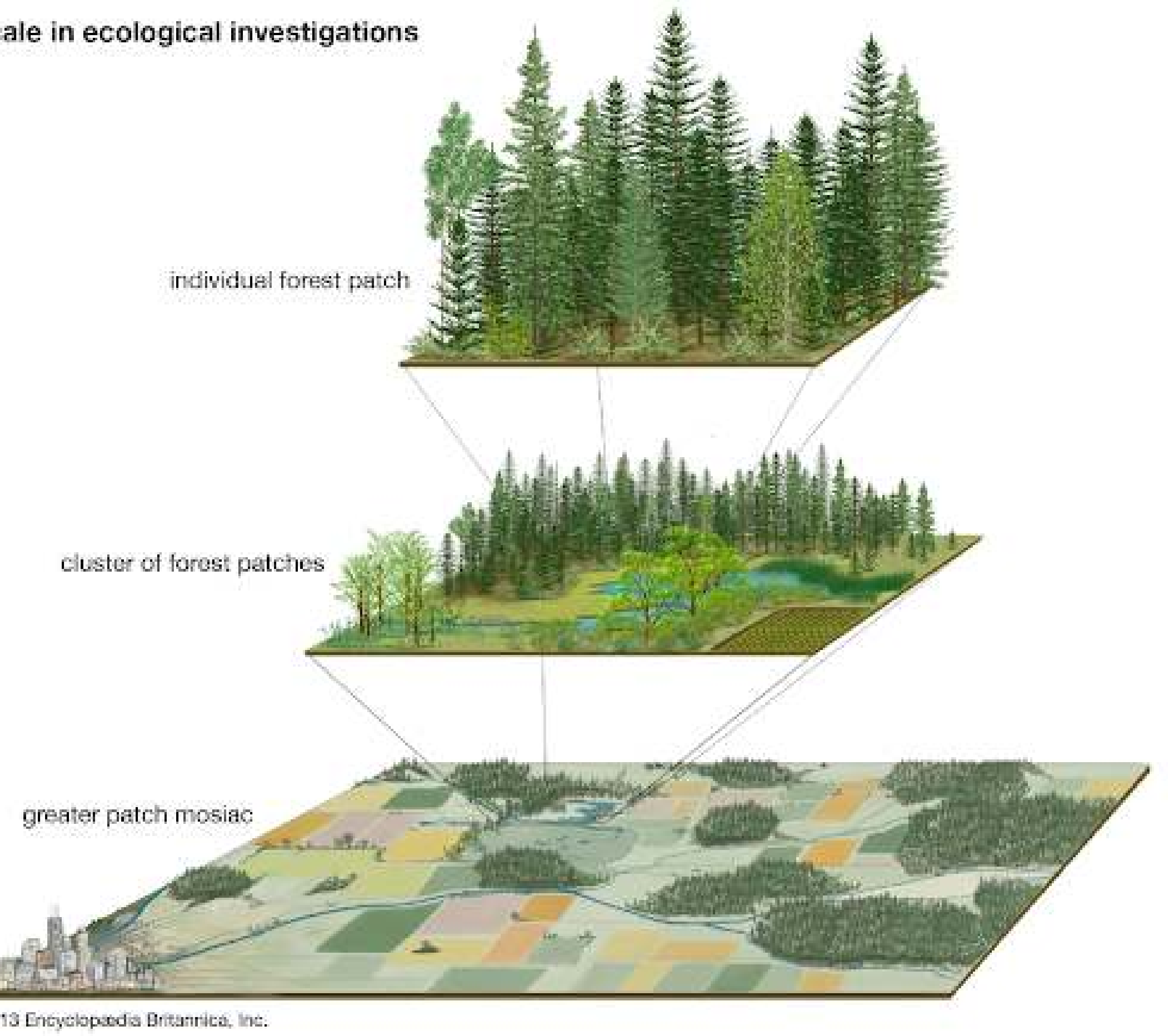
Projections are Political

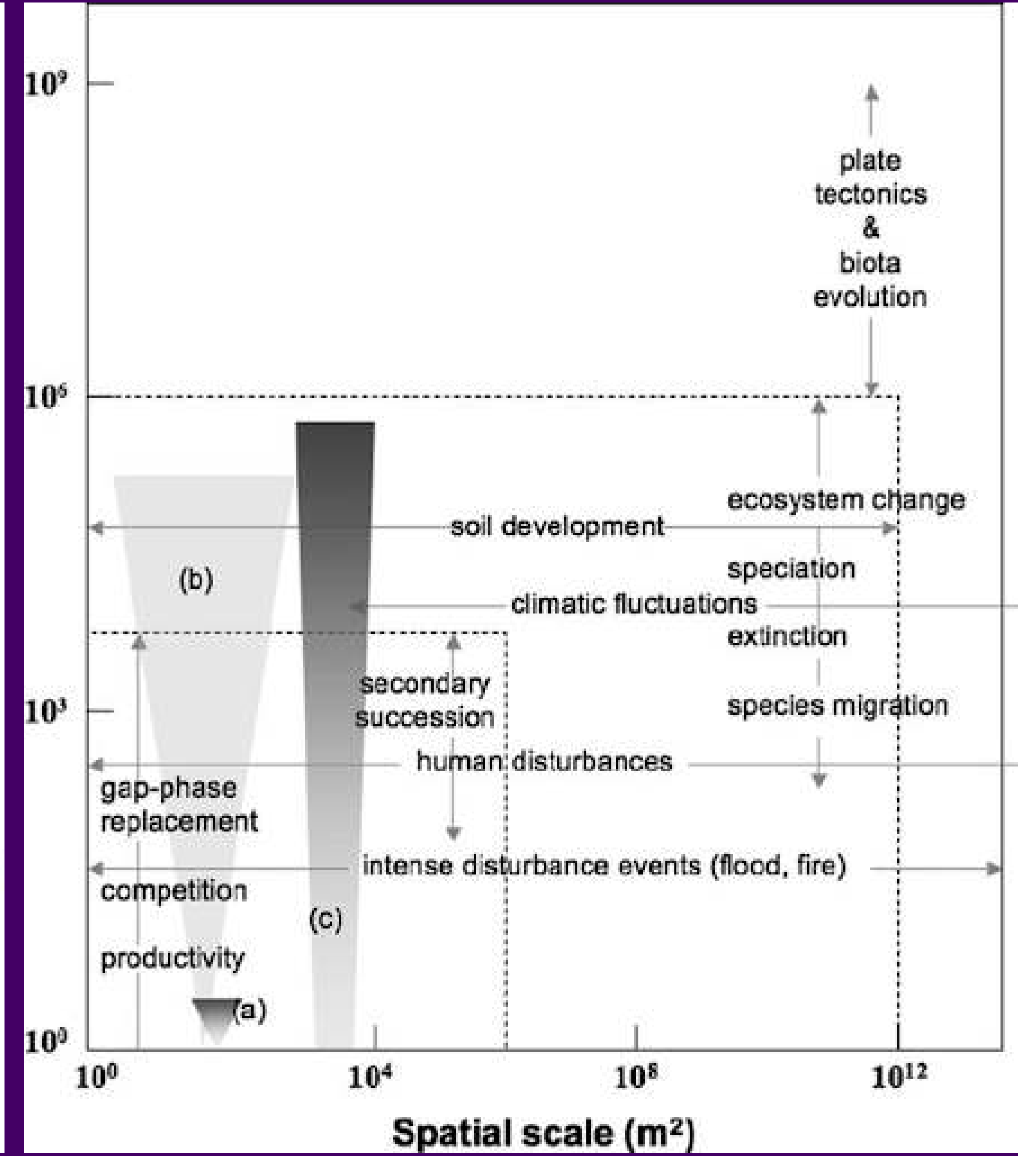
- Convey power through size
- Convey interest through focus
- Convey worldview through orientation

Decolonial Atlas

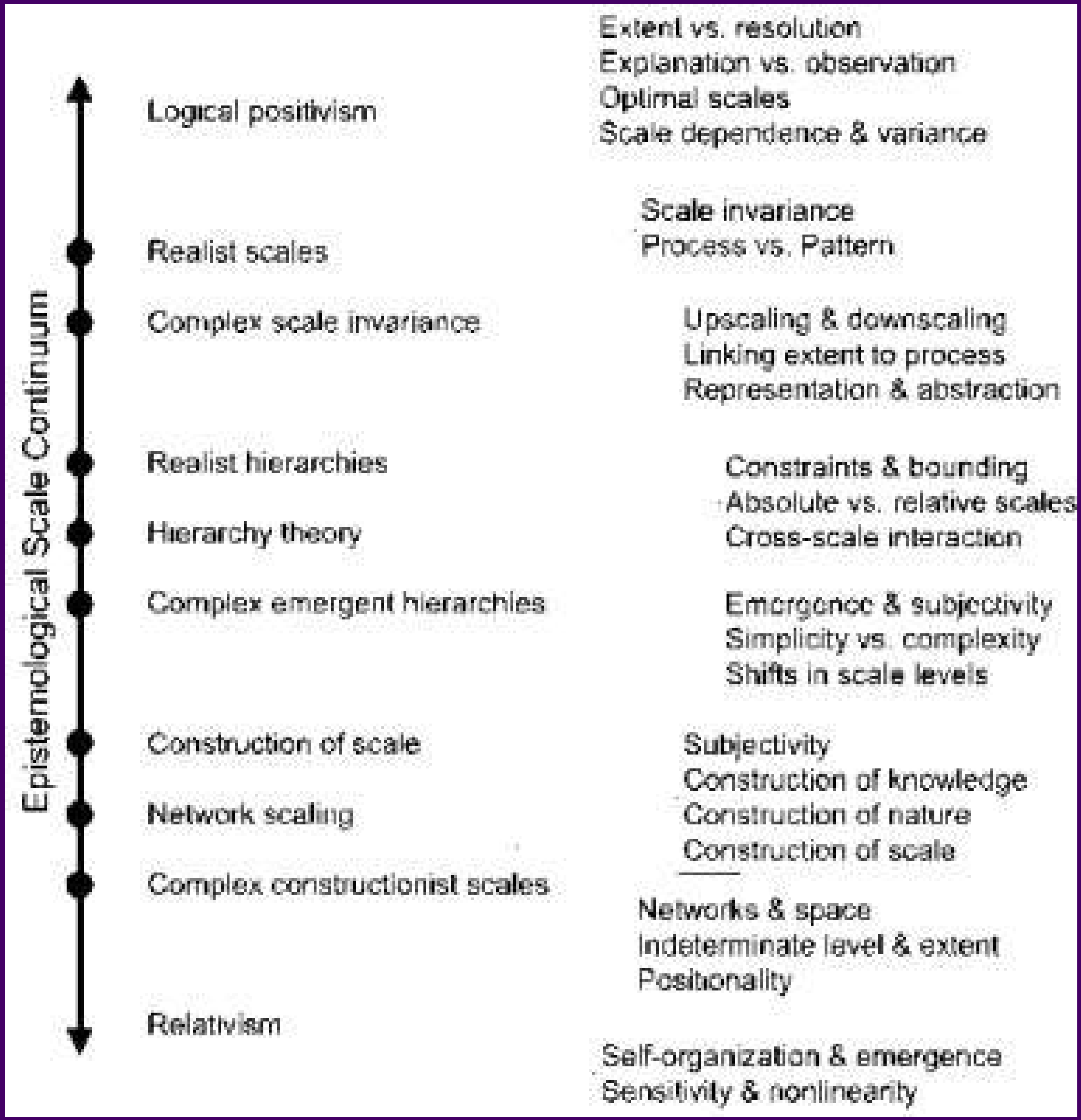
Scale and Geographic Analysis

Matching Inquiry to Process





What do we even mean?

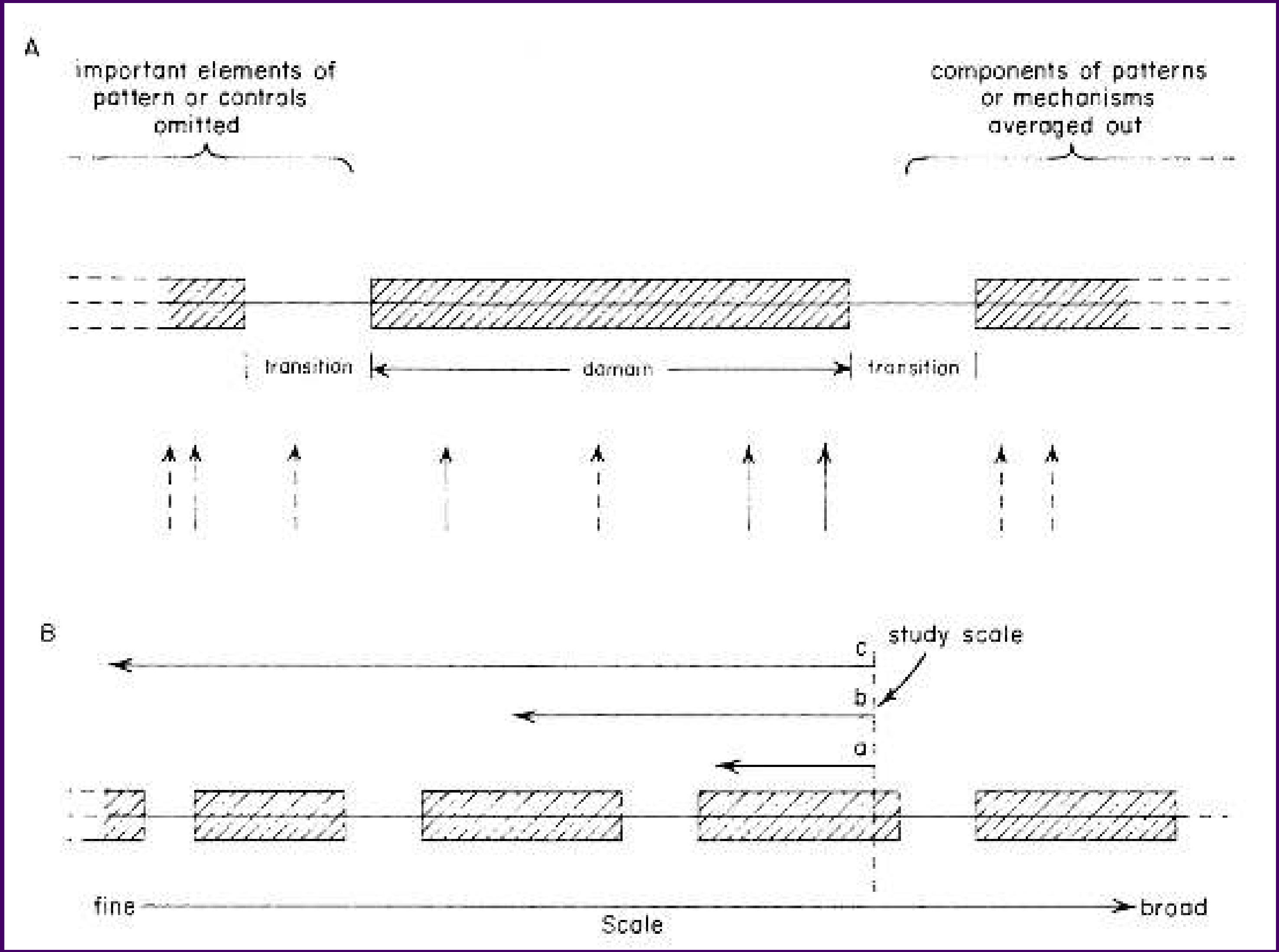


From Manson 2008

- **Grain**: the smallest unit of measurement
- **Extent**: the areal coverage of the measurement

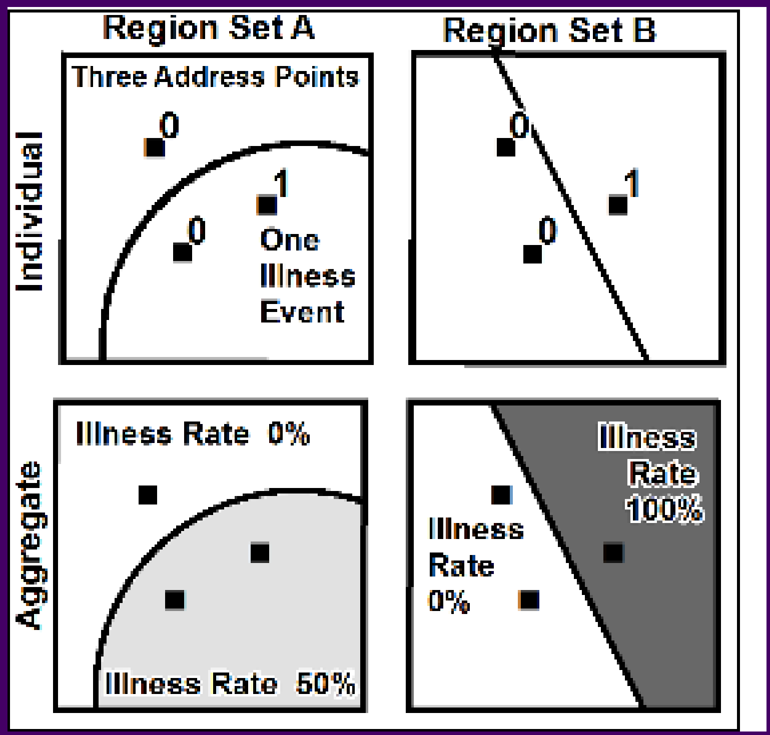
Scale

Even if it exists, how do we know we are measuring at the *right* scale?



Fallacies

- **Locational Fallacy:** Error due to the spatial characterization chosen for elements of study
- **Atomic Fallacy:** Applying conclusions from individuals to entire spatial units
- **Ecological Fallacy:** Applying conclusions from aggregated information to individuals



In Practice

- Who chooses extent?
- How does aggregation affect the data?
- How spatially uniform is the underlying data?
- What can we change?

In Practice

- Projection is not just a visual choice
- Aligning data is critical for robust analysis
- How might your choice of distortion affect your analysis